

Culinary Corridors

Mapping Food Similarity, Spatial Residuals, and Regional Exchange

Full Investigation Report for the Fisher Award Project

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with project mentorship context from Prof. Pia Sorensen

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Abstract

This report consolidates the investigation behind *Culinary Corridors*, a Fisher Award project that treats cuisine as spatial evidence. The central question is whether similarity between cuisines is mostly explained by geographic distance, or whether residual similarities reveal corridors, bridges, regional clusters, and boundary patterns that require geospatial analysis to detect. The project used a prepared What's Cooking/Yummly-derived recipe corpus with cuisine labels and ingredient lists, constructed cuisine-by-ingredient profiles, computed pairwise cuisine similarity, mapped cuisines to coordinate proxies, fitted a distance-only baseline, and analyzed residuals as candidate culinary corridors. The final scope is deliberately conservative: a global 20-cuisine model is used as a discovery screen; East/Southeast Asia is the primary inference case; the Iberian/Atlantic-Pacific pattern is treated as a secondary, diagnostic corridor hypothesis. The key finding is that cuisine similarity is spatially structured, but not reducible to straight-line distance. The most Fisher-relevant insight comes from residual bridge scores and focused corridor maps, which convert food-similarity residuals into mapped spatial roles that could not be obtained from ingredient data alone.

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1 Executive Summary

Project title: *Culinary Corridors: Mapping Food Similarity, Spatial Residuals, and Regional Exchange.*

Research question: When two cuisines are similar, is that similarity mostly explained by geographic distance, or do residual similarities reveal spatial corridors, regional bridges, and boundary patterns that distance alone cannot explain?

Final scope: The project uses a global discovery screen followed by focused interpretation. The global layer uses all 20 cuisines in the corpus to identify broad residual patterns. The strongest focused case is East/Southeast Asia. The Iberian/Atlantic-Pacific case is retained as a secondary, hypothesis-generating case because it is visually compelling but more exposed to platform bias and overinterpretation.

Core finding: Geography explains part of cuisine similarity, but not all of it. In the filtered Run 2 v2 model, cuisine similarity declined with log geographic distance, with a coefficient of -0.1158 and $R^2 = 0.3553$ across 190 dyads. Positive residuals become the main spatial object: they identify cuisine pairs whose observed ingredient-profile similarity is higher than predicted from distance alone.

Most defensible interpretation: The analysis shows spatial structure in cuisine similarity in the prototype dataset. It does *not* prove that migration, trade, colonial history, maritime exchange, or fermentation caused any particular link. Mechanism language should be limited to “candidate corridor,” “spatial association,” or “consistent with,” unless future work adds explicit covariate modeling.

Why this fits Fisher: The project is not simply a map of food. GIS produces the insight by turning ingredient similarity into spatial residuals, focused corridor maps, boundary/permeability summaries, and residual bridge scores. This aligns with the Fisher Award’s emphasis on geographic information science, spatial analysis, interdisciplinary data, and clear geographic communication [1, 2].

2 Background and Motivation

Food is simultaneously cultural, chemical, ecological, historical, and spatial. Ingredients and cooking techniques move through migration, trade, climate zones, ports, borders, households, and markets. A recipe corpus can therefore be used not only as a cultural dataset but also as a spatial dataset: each cuisine label becomes a point or region, each ingredient profile becomes an attribute vector, and each pair of cuisines becomes a dyadic spatial relationship.

The project began with a broad food-science interest: food has global reach, regional variation, shared ingredients, fermentation techniques, and possible links to migration and microbial practices. The investigation narrowed that broad idea into a Fisher-style spatial question. The winning version is not “food around the world.” It is a spatial-inference workflow:

Use cuisine similarity as a signal, compare it to geographic expectation, then map the residuals where food similarity exceeds what distance alone predicts.

This framing gives Prof. Pia Sorensen a clear role as food-science mentor: validating ingredient normalization, advising on whether flavor chemistry should be a secondary layer, identifying overclaims about cooking technique or fermentation, and improving the scientific language around food similarity.

3 The Problem

A global cuisine map can be attractive but analytically weak if it merely shows where foods are located. The project’s challenge is to produce insight that can only come from a geospatial approach.

The main problem is methodological:

How can we distinguish ordinary ingredient similarity from spatially meaningful similarity?

A cuisine pair may be similar because it is geographically close, because it shares climate and staple crops, because recipes on an English-language platform use generic pantry ingredients, because cuisine labels are broad, or because the pair reflects migration, trade, colonial, maritime, or regional exchange. A credible Fisher project must separate these possibilities enough to make a strong but modest claim.

The solution was to treat straight-line geographic distance as a baseline rather than as the final explanation. If cuisine similarity is modeled as a function of distance, then the residuals identify where the food pattern is spatially surprising. Those residuals can then be mapped, aggregated, and compared across regions and boundaries.

4 Investigation Timeline

1. **Run 1: Topic and feasibility.** The project compared cuisine similarity, flavor chemistry, fermentation, and combined culinary-corridor variants. The strongest path was a combined project with cuisine similarity as the core, flavor chemistry as a possible secondary layer, and fermentation deferred unless strong geocoded microbiome data appeared.
2. **Run 2: Real-data prototype.** The project staged and cleaned a prepared What’s Cooking/Yummly-derived recipe corpus, built cuisine-ingredient matrices, computed cosine and Jaccard similarity, created distance baselines, mapped residual corridors, and tested one exploratory overlay.
3. **Run 2 v2: Scope correction and geospatial strengthening.** The project narrowed final inference to a global discovery screen plus focused cases. It added generic-ingredient filtering, East/Southeast Asia and Iberian/Atlantic-Pacific case results, residual bridge scores, path/connectivity proxies, and boundary/permeability checks.
4. **Run 3: Submission packaging.** The final package selected figures, wrote StoryMap-ready prose, prepared a static report, appendices, claim audit, and a Pia review packet. This full report consolidates those findings into LaTeX/PDF form.

5 Data Sources and Scope

5.1 Recipe corpus

The prototype uses a prepared What’s Cooking/Yummly-derived recipe corpus. The corpus is useful because it contains recipe-level cuisine labels and ingredient lists, but it must be interpreted as platform-mediated recipe data rather than a representative census of global cooking practices [5, 6].

Table 1: Core data spine used in the prototype.

Quantity	Value
Recipe records retained	39,774
Recipe-ingredient rows	428,249
Cuisine labels retained	20
Raw ingredient strings observed	6,714
Normalized ingredient labels observed	5,936
Original cuisine-ingredient matrix	20 cuisines $\times$ 1,434 retained ingredients
Filtered sensitivity matrix	20 cuisines $\times$ 1,430 ingredients
Dyadic cuisine pairs modeled	190

The analysis uses coordinate proxies and regional/subregional groupings for cuisine labels. UN M49 was used for regional and subregional classification, and Natural Earth is the appropriate public-domain basemap source for polished mapping [7, 8]. Cuisine labels are not treated as nation-states: labels such as Southern U.S. and Cajun/Creole are regional or diasporic, while broad labels such as Chinese or Indian compress substantial internal diversity.

5.2 Cuisine sample sizes

The corpus contains 20 cuisine labels. Sample sizes are uneven, with Italian, Mexican, and Southern U.S. much larger than Brazilian, Russian, Jamaican, Irish, or Filipino categories. This unevenness reinforces the need for cautious interpretation.

Table 2: Recipe counts by cuisine label.

Cuisine	Recipes
Brazilian	467
British	804
Cajun/Creole	1,546
Chinese	2,673
Filipino	755
French	2,646
Greek	1,175
Indian	3,003
Irish	667
Italian	7,838
Jamaican	526
Japanese	1,423
Korean	830
Mexican	6,438
Moroccan	821
Russian	489
Southern U.S.	4,320
Spanish	989
Thai	1,539
Vietnamese	825

5.3 Generic-ingredient filtering

Generic ingredients can inflate similarity between distant cuisines. Run 2 v2 introduced a reproducible policy that removed or downweighted generic pantry terms and high-prevalence ingredients rather than silently deleting them. The policy was important because global residual rankings changed after filtering.

6 Methodological Approach

6.1 Cuisine vectors

Each cuisine c is represented by an ingredient vector $\mathbf{x}_c$, where vector entries represent normalized ingredient frequencies or prevalence weights. The cuisine-by-ingredient matrix is the basic food-data object.

6.2 Similarity metrics

The main similarity metric is cosine similarity:

$$s_{ij} = \cos(\mathbf{x}_i, \mathbf{x}_j) = \frac{\mathbf{x}_i \cdot \mathbf{x}_j}{\|\mathbf{x}_i\| \|\mathbf{x}_j\|}. \quad (1)$$

Jaccard similarity was used as a robustness metric after thresholding ingredient presence. Cosine similarity was used for the distance-residual model because it naturally compares weighted cuisine vectors.

6.3 Cuisine-to-place mapping

Each cuisine label was assigned a coordinate proxy with confidence notes. The mapping is necessary for distance calculation and corridor visualization, but it is also a source of uncertainty. Therefore, the report avoids precise origin claims and uses coordinates as analytical proxies.

6.4 Distance baseline

The central model asks whether cuisine similarity declines with geographic distance:

$$s_{ij} = \alpha + \beta \log(d_{ij}) + \epsilon_{ij}, \quad (2)$$

where s_{ij} is cosine similarity and d_{ij} is great-circle distance in kilometers between cuisine-coordinate proxies.

The residual is the project's key spatial object:

$$r_{ij} = s_{ij} - \hat{s}_{ij}. \quad (3)$$

A positive residual indicates that two cuisines are more similar than a distance-only model predicts. A negative residual indicates that they are less similar than expected given distance.

6.5 Focused cases

The global model is used only as a discovery screen. Final inference is focused on:

- **East/Southeast Asia:** Chinese, Japanese, Korean, Thai, Vietnamese, Filipino.
- **Iberian/Atlantic-Pacific:** Spanish, Brazilian, Mexican, Filipino, Cajun/Creole, Jamaican, Southern U.S.

The first case is the strongest because it is spatially coherent and has multiple positive residuals after filtering. The second is visually useful but more exposed to platform bias and coarse historical interpretation.

6.6 Geospatial-only analysis

Ingredient vectors can identify similarity, but they cannot identify mapped spatial roles. Run 2 v2 therefore added residual bridge scores. A bridge score aggregates positive residual links into place-level roles:

$$B_i = \sum_{j:r_{ij}>0, d_{ij}>d_0} r_{ij}, \quad (4)$$

where d_0 marks long-distance residual links in the prototype. This score depends on residuals, coordinates, and distance; it is therefore a geospatial measure rather than merely an ingredient-clustering measure.

7 Final Figures and Findings

7.1 Finding 1: global residuals are useful for discovery, not proof

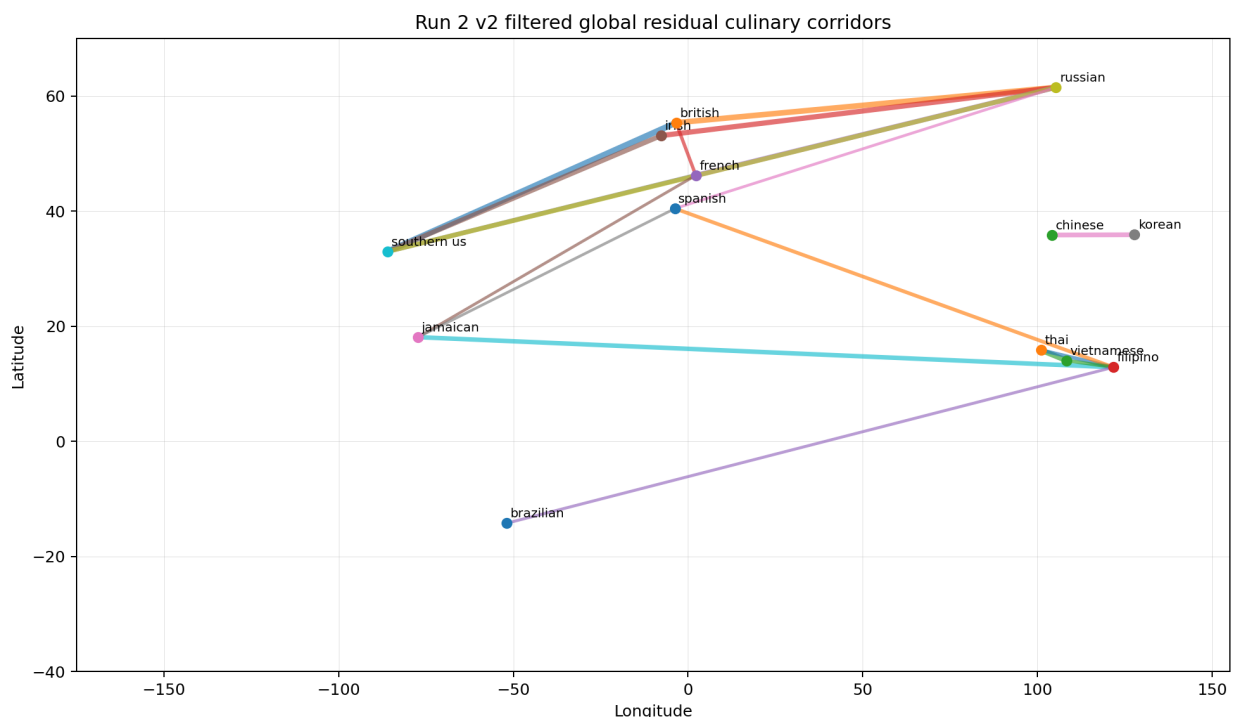


Figure 1: Global discovery: filtered residual culinary corridors. Lines connect cuisine pairs whose observed ingredient-profile similarity is higher than a distance-only model predicts after generic-ingredient filtering. This is a discovery screen, not proof of migration, trade, colonial, or maritime causation.

Figure 1 shows the final filtered residual corridor map. It is valuable because it locates candidate corridors, but it is intentionally not the final causal argument. Long-distance residuals can arise from true spatial history, shared ingredient ecology, broad cuisine labels, generic recipe-platform vocabulary, or platform bias.

7.2 Finding 2: distance matters, but it is incomplete

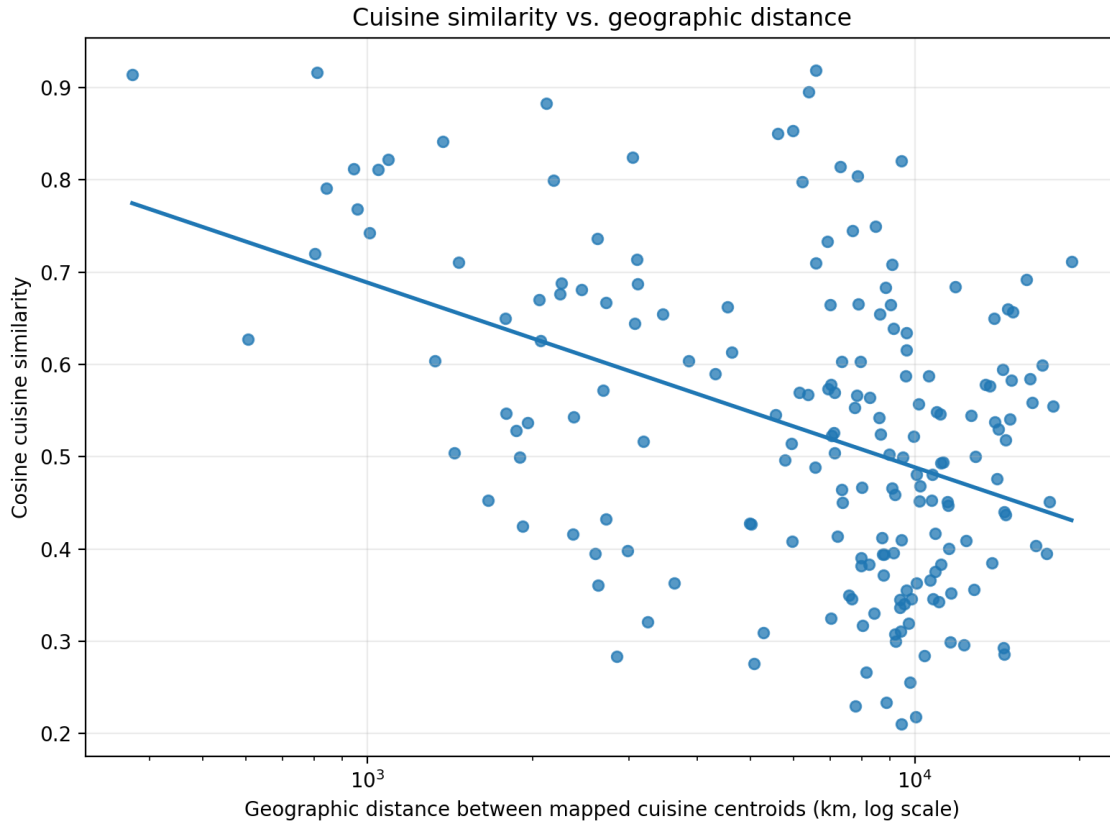


Figure 2: Distance baseline and residual logic. The Run 2 v2 filtered model reports a negative log-distance coefficient and $R^2 = 0.3553$, meaning that geographic distance explains part of cuisine similarity but leaves residual structure to analyze.

The filtered model estimated:

$$\hat{s}_{ij} = 1.2727 - 0.1158 \log(d_{ij}),$$

$$R^2 = 0.3553,$$

$$N = 190 \text{ dyads.}$$

This is a strong prototype result because it gives the project a spatial baseline without overclaiming that distance explains everything.

Table 3: Top filtered positive residual culinary corridors.

Cuisine pair	Cosine	Jaccard	Distance km	Predicted	Residual
British – Southern U.S.	0.663	0.473	6,599	0.254	0.408
British – Russian	0.644	0.508	5,614	0.273	0.371
Thai – Vietnamese	0.856	0.608	808	0.498	0.359
Irish – Russian	0.607	0.483	5,975	0.266	0.341
French – Russian	0.591	0.471	6,221	0.261	0.330
Irish – Southern U.S.	0.575	0.467	6,390	0.258	0.317
Chinese – Korean	0.692	0.508	2,118	0.386	0.306
French – Southern U.S.	0.544	0.680	7,307	0.243	0.302
Russian – Southern U.S.	0.512	0.408	9,455	0.213	0.300
Filipino – Jamaican	0.436	0.433	15,984	0.152	0.284
Filipino – Thai	0.598	0.466	2,262	0.378	0.219
Filipino – Spanish	0.405	0.368	11,857	0.187	0.219

The top residual table illustrates why the global layer is discovery only. Some links, such as Thai–Vietnamese and Chinese–Korean, are spatially coherent and plausible for focused interpretation. Others, especially Anglo/Southern U.S./European links, may reflect platform vocabulary, baking/pantry ingredients, or English-language recipe patterns and should be treated as diagnostic rather than central.

7.3 Finding 3: East/Southeast Asia is the strongest focused case

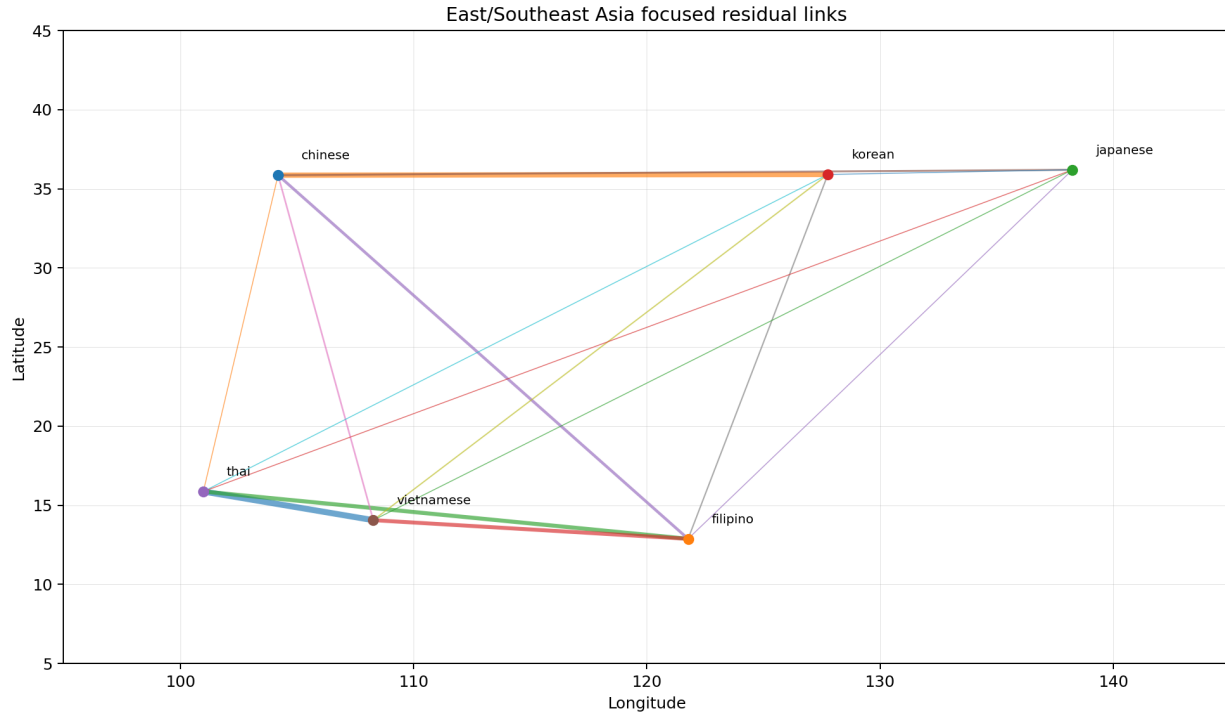


Figure 3: Primary focused case: East/Southeast Asia. This map focuses on Chinese, Japanese, Korean, Thai, Vietnamese, and Filipino cuisines and highlights positive residual links after filtering. It supports spatial association, not a specific causal diffusion route.

The East/Southeast Asia case includes six cuisines and 15 cuisine pairs. Nine pairs have positive residuals. The mean residual is 0.0694 and the median residual is 0.0311, with mean distance of 2,593.7 km across the focused dyads.

Table 4: Top East/Southeast Asia focused-case residual links.

Cuisine pair	Cosine	Jaccard	Distance km	Residual	Spatial relation
Thai – Vietnamese	0.856	0.608	808	0.359	same subregion
Chinese – Korean	0.692	0.508	2,118	0.306	same subregion
Filipino – Thai	0.598	0.466	2,262	0.219	same subregion
Filipino – Vietnamese	0.637	0.474	1,465	0.209	same subregion
Chinese – Filipino	0.475	0.490	3,103	0.133	same region, cross subregion
Chinese – Japanese	0.412	0.607	3,047	0.068	same subregion
Chinese – Vietnamese	0.417	0.583	2,458	0.049	same region, cross subregion
Filipino – Korean	0.392	0.437	2,630	0.031	same region, cross subregion
Korean – Vietnamese	0.366	0.490	3,111	0.024	same region, cross subregion

The strongest focused links are Thai–Vietnamese, Chinese–Korean, Filipino–Thai, Filipino–Vietnamese, and Chinese–Filipino. These results support the modest claim that, within this subset, cuisine similar-

ity remains spatially organized around same-subregion and cross-subregion relationships after generic-ingredient filtering.

7.4 Finding 4: residual bridge scores provide the strongest geospatial-only insight

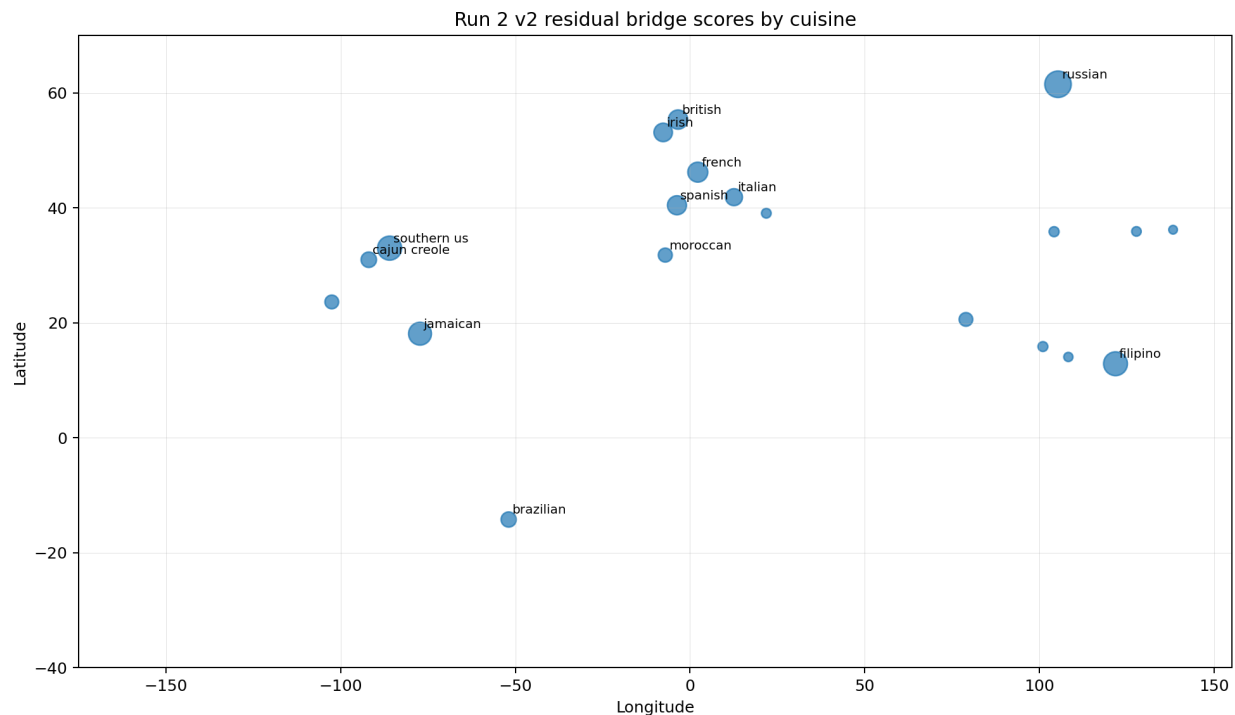


Figure 4: Residual bridge scores. Point size represents long-distance positive residual bridge score. This transforms pairwise residual corridors into mapped place-level roles and is the strongest geospatial-only contribution of the project.

Residual bridge scores ask which cuisine labels repeatedly participate in positive residual links. This requires spatial structure: distance, coordinates, residuals, and pair aggregation. It cannot be obtained from ingredient vectors alone.

Table 5: Top residual bridge-score cuisines.

Cuisine	Case role	+ residual degree	Top-20 degree	Long-distance degree	Long-distance score	Mean residual
Russian	bias diagnostic	13	6	13	2.062	0.051
Southern U.S.	secondary + bias diagnostic	7	4	7	1.661	0.048
Filipino	primary + secondary	17	6	13	1.630	0.110
Jamaican	secondary	15	4	14	1.476	0.069
French	global screen only	11	5	7	1.066	0.050
British	bias diagnostic	8	3	6	0.977	0.010
Spanish	secondary	12	3	8	0.946	0.039
Irish	bias diagnostic	7	2	5	0.879	0.003
Italian	global screen only	9	0	7	0.701	0.015
Cajun/Creole	secondary	9	0	9	0.538	-0.013

The bridge-score results are not all equally interpretable. Filipino is especially interesting because it appears in both the East/Southeast Asia primary case and the Iberian/Atlantic-Pacific secondary case.

Russian, British, Irish, and Southern U.S. patterns are useful as bias diagnostics and should not anchor the final historical claim without stronger external covariates.

7.5 Finding 5: the Iberian/Atlantic-Pacific case is compelling but riskier

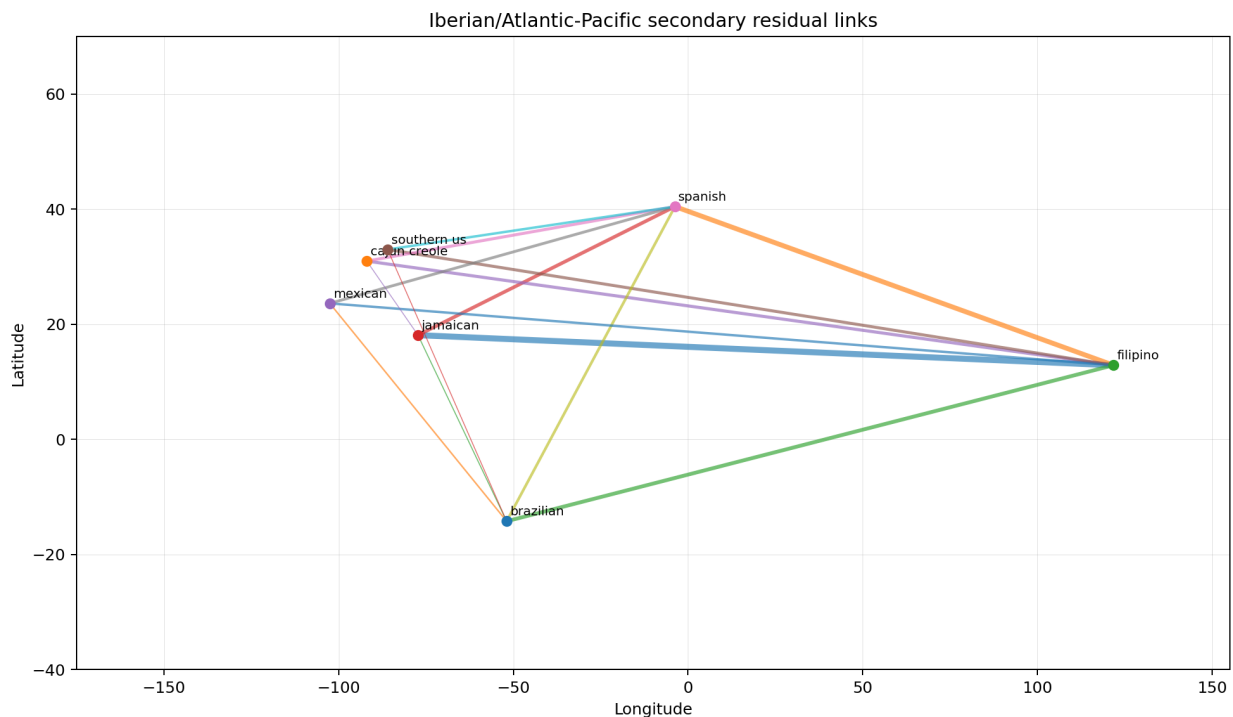


Figure 5: Secondary/diagnostic case: Iberian/Atlantic-Pacific corridor. This case illustrates long-distance residual corridor hypotheses, but it is more vulnerable to platform bias, broad cuisine labels, and overinterpretation of colonial or maritime histories.

The secondary case has 7 cuisines, 21 pairs, and 13 positive residual pairs. Its mean residual is 0.0403 and median residual is 0.0831, with mean distance of 7,757 km. This case should be framed as a hypothesis-generating contrast rather than as proof of historical exchange.

Table 6: Top Iberian/Atlantic-Pacific secondary-case residual links.

Cuisine pair	Cosine	Jaccard	Distance km	Residual
Filipino – Jamaican	0.436	0.433	15,984	0.284
Filipino – Spanish	0.405	0.368	11,857	0.219
Brazilian – Filipino	0.290	0.420	19,318	0.160
Jamaican – Spanish	0.385	0.425	7,340	0.143
Cajun/Creole – Filipino	0.289	0.408	13,951	0.122
Filipino – Southern U.S.	0.281	0.398	14,123	0.115
Cajun/Creole – Spanish	0.350	0.541	7,699	0.114
Mexican – Spanish	0.323	0.574	9,026	0.105
Brazilian – Spanish	0.331	0.435	7,850	0.097
Southern U.S. – Spanish	0.331	0.558	7,109	0.085

7.6 Finding 6: residual strength varies by spatial grouping

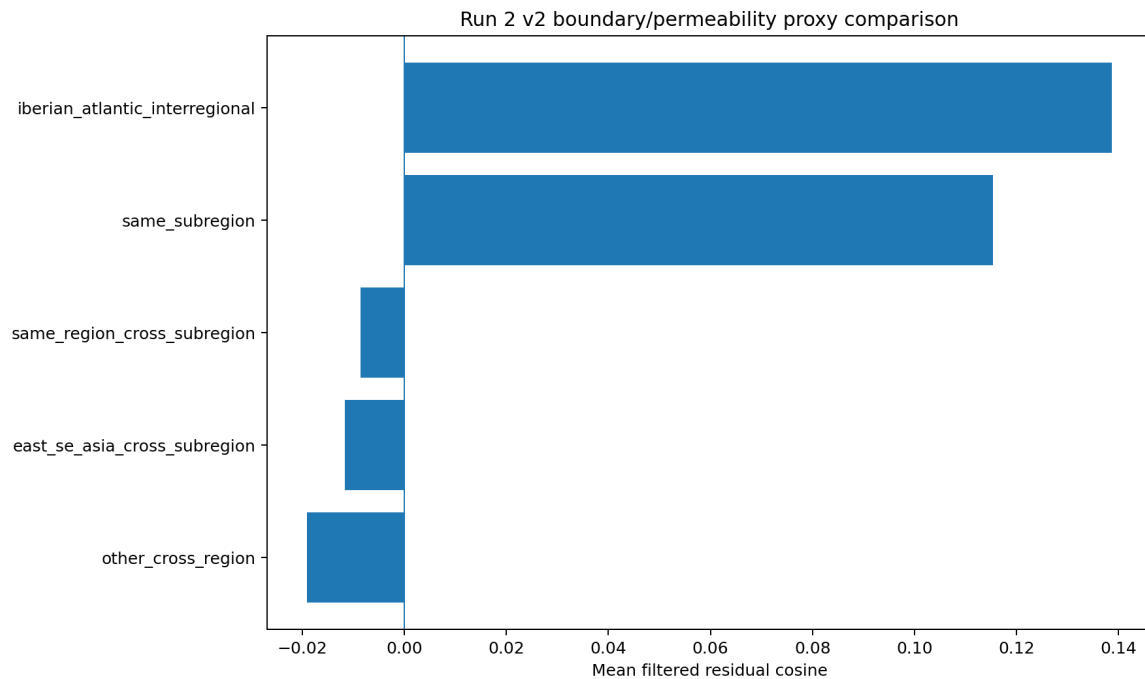


Figure 6: Boundary/permeability proxy. Mean residuals differ across spatial boundary classes, supporting the claim that residual cuisine similarity has spatial structure. This analysis is exploratory and should motivate stronger route or boundary modeling rather than replace it.

Table 7: Boundary/permeability summary.

Boundary class	Pairs	Mean resid.	Median resid.	Mean sim.	Mean dist. km	+ share
Iberian/Atlantic interregional	11	0.139	0.115	0.334	11,634	100.0%
Same subregion	11	0.115	0.129	0.565	1,456	72.7%
Same region, cross subregion	32	-0.009	-0.037	0.341	3,543	37.5%
East/Southeast Asia cross subregion	9	-0.012	-0.006	0.331	3,140	44.4%
Other cross region	127	-0.019	-0.039	0.201	9,675	38.0%

Same-subregion dyads and the Iberian/Atlantic interregional class have higher mean residuals in the prototype. However, the Iberian/Atlantic class should be interpreted with care because it is long-distance, historically tempting, and vulnerable to platform-mediated similarities.

8 Additional Analysis: Spatial Connectivity Proxy

The project also tested a transparent path/connectivity proxy using coordinates, UN region/subregion labels, distance classes, and maritime-exposure flags. This is not a true least-cost path model or historical route model. Its purpose is to show that corridor results can be grouped by spatial accessibility classes and to motivate future route-aware GIS.

Table 8: Path/connectivity proxy summary.

Connectivity proxy class	Pairs	Mean residual	Mean distance km	+ share
Same subregion	11	0.115	1,456	72.7%
Long-distance cross region	21	0.023	9,691	61.9%
Long-distance maritime proxy	107	-0.003	10,522	41.1%
Same region, cross subregion	41	-0.009	3,454	39.0%
Other cross region	10	-0.099	2,734	20.0%

The strongest substantive next step would be to replace this proxy with actual route-aware or cost-distance data: port networks, historical maritime routes, rail/road accessibility, ecological barriers, or least-cost surfaces. ArcGIS-style cost-distance and network approaches would be appropriate for a more advanced version of the project.

9 Interpretation and Claim Discipline

9.1 Strong conclusions

The project can strongly conclude the following:

- Cuisine similarity is spatially structured in the prototype dataset.
- Straight-line geographic distance explains some, but not all, cuisine similarity.
- Global residual maps are useful discovery tools but are too broad for causal claims.
- East/Southeast Asia is the strongest focused case in the current dataset.
- Residual bridge scores provide a Fisher-style geospatial insight because they convert residual similarity into mapped place-level roles.

9.2 Cautious hypotheses

The project can cautiously suggest:

- Some positive residual corridors may be consistent with migration, trade, colonial, maritime, or agricultural exchange histories.
- East/Southeast Asian residuals may reflect regional adjacency, shared ingredient ecologies, food technologies, and land/sea exchange zones.
- Iberian/Atlantic-Pacific links may be consistent with long-distance historical exchange, but require explicit covariates and stronger data before being interpreted historically.

9.3 Forbidden claims

The project should not claim:

- that the recipe corpus represents all world cuisines;
- that residual links prove migration, trade, empire, colonialism, maritime exchange, or diffusion;
- that cuisine labels are exact nation-state units;
- that flavor chemistry or fermentation explains the final results;
- that the global residual map is a causal model.

10 Role of Prof. Pia Sorensen

Pia's most valuable role is to improve the food-science validity of the interpretation rather than to serve as a general project reviewer. The report recommends asking her to assess:

1. whether the generic-ingredient filtering policy is scientifically defensible;
2. whether ingredient groups should be merged, split, or treated as functionally different;
3. whether East/Southeast Asian links are better explained through ingredient ecology, technique, preparation, fermentation, or flavor structure;
4. whether flavor chemistry should be included as a sidebar or deferred;
5. which food-science language would prevent overclaiming from recipe-derived ingredient data.

If Pia feedback is unavailable, the final text should state that food-science interpretation is provisional.

11 Limitations

1. **Platform bias.** The recipe corpus likely reflects online, English-language, and platform-specific categories rather than global cooking practice.
2. **Broad cuisine labels.** Cuisines such as Chinese, Indian, and Mexican contain large regional diversity; labels such as Southern U.S. or Cajun/Creole are not countries.
3. **Coordinate proxies.** Coordinates approximate cuisine labels and cannot represent diasporic, regional, or historical food geographies precisely.

4. **Ingredient normalization.** Rule-based normalization may merge or separate ingredients in ways that affect similarity.
5. **Generic ingredients.** Pantry ingredients strongly affect similarity. Filtering reduces this problem but involves interpretive choices.
6. **Residuals are not mechanisms.** A positive residual is evidence of unexpected similarity relative to distance. It is not evidence of a specific historical cause.
7. **Flavor chemistry and fermentation.** These remain future or mentor-validated layers, not current explanatory mechanisms.

12 Fisher Award Fit

The Fisher Award context favors projects where GIS analysis is central rather than decorative. This project satisfies that criterion in several ways:

- The map is not merely illustrative; distance modeling creates the residual object to be mapped.
- Focused maps separate discovery from inference, improving claim discipline.
- Residual bridge scores use coordinates, distance, and residuals to identify mapped spatial roles.
- Boundary/permeability and connectivity classes add explicitly spatial structure.
- The project is interdisciplinary: GIS, food science, cultural geography, recipe data, and spatial statistics.

The strongest Fisher line is: *this project uses GIS to show where cuisine follows geography and where it breaks geography.*

13 Recommendations for Final Submission

1. Use an ArcGIS StoryMap or web-map-heavy format if available. The narrative is naturally visual and spatial.
2. Lead with the global discovery figure, but immediately explain that it is not proof.
3. Make the East/Southeast Asia case the main inference section.
4. Use residual bridge scores as the key Fisher differentiator.
5. Include the Iberian/Atlantic-Pacific case only as a secondary, diagnostic, or hypothesis-generating case.
6. Keep flavor chemistry as future work unless Pia validates the matching strategy.
7. Put platform bias, cuisine-label uncertainty, and non-causal interpretation in the main text, not just the appendix.

A Appendix A: Similarity Heatmap from Run 2

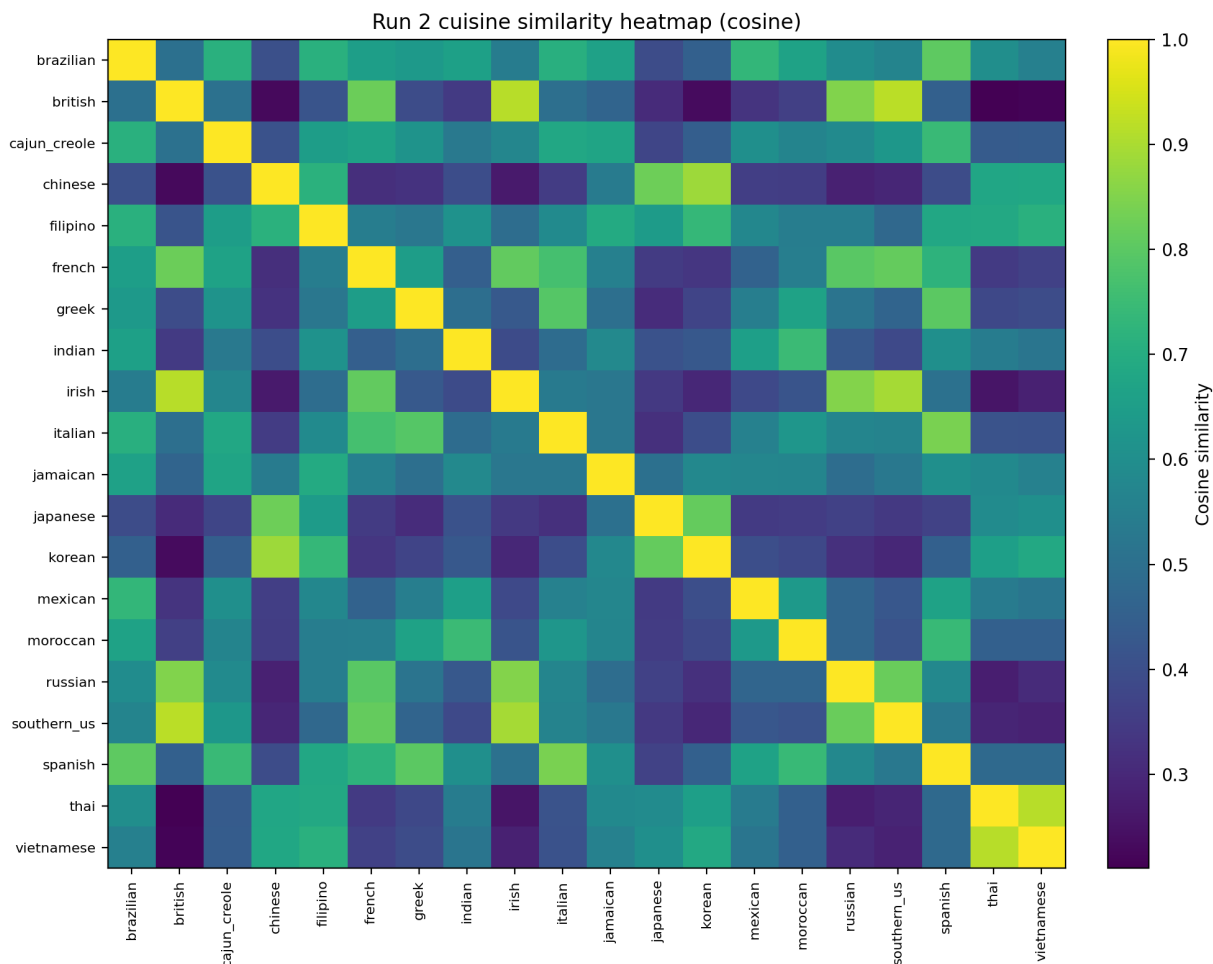


Figure 7: Run 2 cuisine similarity heatmap. This figure is useful for showing ingredient-space clustering, but it was demoted from the main final narrative because the final Fisher story is about spatial residuals rather than clustering alone.

B Appendix B: Technical Reproducibility Summary

The final report reuses Run 2 and Run 2 v2 outputs. Key scripts include:

- `scripts/02_clean_recipe_ingredients.py`: ingredient cleaning and long-table construction;
- `scripts/03_build_cuisine_matrix.py`: cuisine-by-ingredient matrix;
- `scripts/04_compute_similarity.py`: cosine and Jaccard similarity;
- `scripts/05_build_distance_pairs.py`: pairwise distance table;
- `scripts/07_fit_distance_baseline.py`: distance-only model and residuals;
- `scripts/10_run2v2_filter_ingredient_matrix.py`: generic-ingredient sensitivity model;
- `scripts/12_run2v2_focus_case_models.py`: focused cases;

- `scripts/14_run2v2_residual_bridge_scores.py`: residual bridge scores;
- `scripts/15_run2v2_boundary_permeability_check.py`: boundary/permeability summary.

C Appendix C: Source and Citation Notes

Public-facing claims should cite public sources for the Fisher Award, dataset references, GIS classifications, and map layers. Internal artifacts should be used for reproducibility but not as substitutes for public source citation where public claims are made.

References

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